Project Report

1. Project Title

Gear Train & Power Transmission Calculator

2. Project Objective

The purpose of this project was to better understand how mechanical systems transfer motion and power through gear mechanisms.

By combining mathematics, physics, and programming, I wanted to create a simple engineering tool that demonstrates how gear ratios influence rotational speed, torque, and overall system performance.

3. Research Question

How do gear ratios affect rotational speed (RPM), torque, and power transmission in mechanical systems?

4. Background

Gears are among the most widely used mechanical components in engineering. They are found in automobiles, aircraft systems, industrial machinery, and countless other applications.

Although gears have appeared simple, they play a critical role in controlling speed and torque. By changing gear ratios, engineers can design systems that prioritize either speed or force depending on the desired application.

This project explores these relationships through mathematical calculations and engineering analysis.

5. Methodology

The calculator was developed using Python. The user enters:

- Driver Gear Teeth
- Driven Gear Teeth
- Input RPM
- Input Torque

The program then calculates:

- Gear Ratio
- Output RPM
- Output Torque

The calculations are based on standard mechanical engineering relationships.

Gear Ratio
Gear Ratio = Teeth Driven / Teeth Driver
Output RPM
Output RPM = Input RPM / Gear Ratio Output Torque
Output Torque = Input Torque × Gear Ratio

6. Results

The calculator successfully analyzes the effect of gear ratios on speed and torque.

Example:

Example:
Driver Gear Teeth = 12
Driven Gear Teeth = 48
Input RPM = 1200 RPM
Input Torque = 10 Nm

Results:

✅ Gear Ratio = 4:1
✅ Output RPM = 300 RPM
✅ Output Torque = 40 Nm

Engineering Interpretation:

✅ The gear train increases torque by a factor of 4 while reducing rotational speed by a factor of 4, demonstrating mechanical advantage through gear reduction.

The simulation demonstrates how increasing gear ratio decreases rotational speed while increasing torque**.**

7. Engineering Insight

The results demonstrate the fundamental trade-off between speed and torque in mechanical systems. Increasing gear ratio provides greater mechanical advantage while reducing output rotational speed, allowing engineers to optimize systems for different performance requirements.

8. Skills Developed

- Python Programming
- Mechanical Engineering Fundamentals
- Engineering Calculations
- Mathematical Modeling
- System Thinking
- Data Analysis
- Problem Solving
- Critical Thinking

9. What I Learned

One of the most interesting things I learned from this project is that small mechanical changes can have a Major impact on system performance.

Before working on this calculator, I understood the basic idea of gears, but I did not fully appreciate how strongly gear ratios affect speed and torque.

Building this project helped me better understand how engineers use mathematics to design machines, improve performance, and solve practical problems.

I also learned how programming can be used to model mechanical systems and visualize engineering concepts.

10. Future Improvements

Possible future developments include:

- Multi-Stage Gear Systems
- Gear Efficiency Calculations
- Power Loss Analysis
- Interactive Mechanical Dashboard
- Mechanical System Visualization
- Real-World Vehicle Transmission Simulation

11. Engineering Significance

Understanding power transmission systems is essential in mechanical Engineering

because gears are widely used to control force, speed, and efficiency in machines.

This project demonstrates how mathematical modeling and programming can be

applied to analyze real-world engineering systems.

12. Personal Reflection

Before this project, I viewed gears mainly as simple machine components.

Learning about gear ratios, RPM, torque, and power transmission showed me how essential gears are to engineering design. Even a simple mechanism can significantly change how a machine operates.

The most rewarding part of the project was seeing how a few mathematical equations could explain the behavior of real mechanical systems.

It reminded me that engineering involves not only building machines but also understanding how systems work and applying simple ideas to solve complex problems.

Engineering is the art of turning motion into practical solutions.